Lin 2020 PDAC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates 15 samples (iLISI 1.0→4.1, same-sample frac 0.94→0.45); one continuous connected component after removing a justified detached island (652 cells, 14 samples, developmentally flat).

Gate 2 — module geography → GO (non-branching)

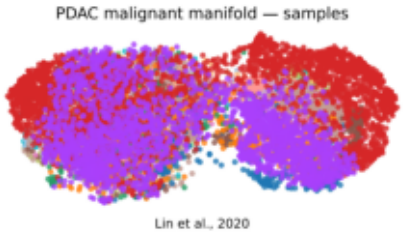
Single pSom→Crest axis: pSom high over ductal body falls along pseudotime (ρ=−0.59), Crest rises to a terminal end (ρ=+0.32, decile pt 0.82); anti-correlated in space (r=−0.38, joint-high enrichment 0.52 = no hub). Gast weak (ρ=−0.24). Crest terminal, not on the Gast end → no inversion flag.

Gate 3 — cross-repression → NA

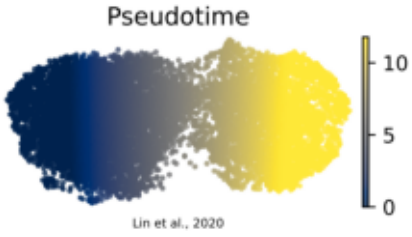
No bifurcation (branching-only gate). Geometry is linear, so the tree cross-repression test does not apply.

STEP FIGURES

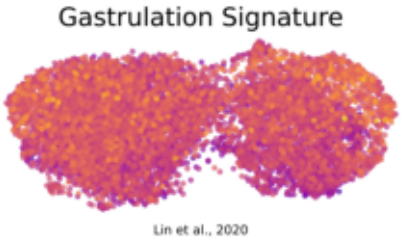
Manifold — samples



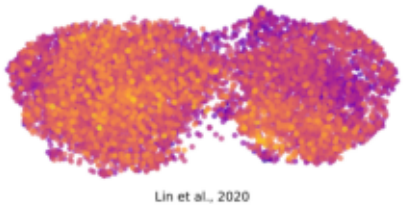
Pseudotime



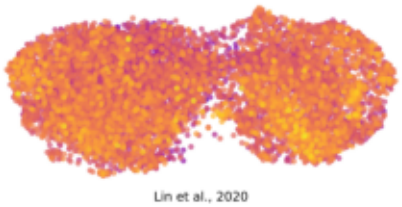
Gastrulation



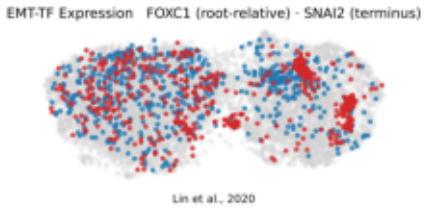
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic Ductal Adenocarcinoma (Lin et al., 2020), 10x — 16 samples; 10,993 malignant cells (10,304 after threshold + island removal) enter the analysis; the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic pattern reproduces as a single somitogenesis→neural-crest developmental axis (the non-branching / collapsed-bifurcation case) — no Gast/pSom hub, no competing arms; Linear pSom/Crest. Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. EMT-TFs FOXC1 (root-relative) → SNAI2 (terminus).

Parameter choices

Threshold	53 cells (geom-mean 76→37, gap ratio 2.05) → 15/16 kept	**Island**	652 cells removed (radius-graph 6× med-NN; multi-sample, dev-flat)
Integration	Harmony θ=10; n_pcs=20; spread=1.3	**Trajectory**	scFates curve (ElPiGraph) Nodes=20, μ=100 (2 tips, 0 forks; self-approach 2.98)
Root	pSom-high / Crest-low tip	**Scope**	whole manifold kept (modules cycle-free; axis holds in non-cycling lobe)
EMT-TF axis	FOXC1 (root-relative) → SNAI2 (terminus)	**Caveat**	pt vs cycling ρ=+0.42 (cycle-split lobes)